

# Stock Market Analysis in SQL

## Market Insights Report

6 NSE stocks | 01 Jan 2015 to 31 Jul 2018 | 889 trading days | bonus-adjusted prices | risk-free rate 6.0% | generated 30 Sep 2026

### Executive summary

TVS Motors led on growth (+19.1% CAGR) while TVS Motors was the most volatile; 3 of 6 stocks show a Buy signal and average correlation is 0.24.

### Key insights

#### PERFORMANCE

##### TVS Motors led the period

TVS Motors: +86.9% total, +19.1% CAGR. Hero Motocorp trailed at +6.0% total.

#### RISK

##### TVS Motors carried the most volatility

TVS Motors: 30.8% annualised volatility. Deepest drawdown: TVS Motors (-34.6%). Best Sharpe (rf 6.0%): TVS Motors (0.43).

#### VOLATILITY CHANGE

##### Bajaj Auto is the most unsettled right now

Last-60-day volatility is 1.36x its full-period level for Bajaj Auto, and 0.74x for Eicher Motors (calmest).

#### TREND

##### 3 of 6 stocks sit on a golden-cross Buy

Buy: TCS, Infosys, Bajaj Auto. RSI above 70: none. RSI below 30: none.

#### DIVERSIFICATION

##### Average pairwise correlation is 0.24

Closest pair: Bajaj Auto / Hero Motocorp (0.52). Most diversifying: Eicher Motors / TCS (0.10). Same-sector pairs average 0.36 vs 0.14 across sectors, consistent with sector-level behaviour in this sample.

#### DATA QUALITY

##### Bonus issues detected in the raw feed

Infosys, TCS show >25% one-day price gaps in raw prices. Everything here uses bonus-adjusted prices, so returns and signals are not distorted.

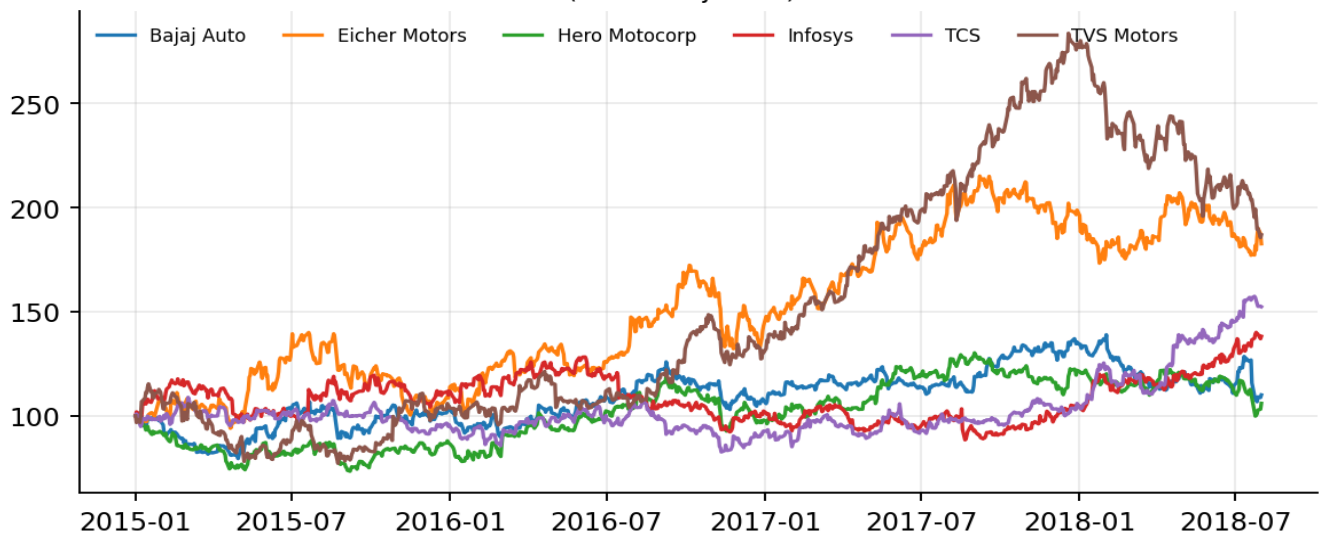
#### BUSINESS

##### Different objectives point to different names

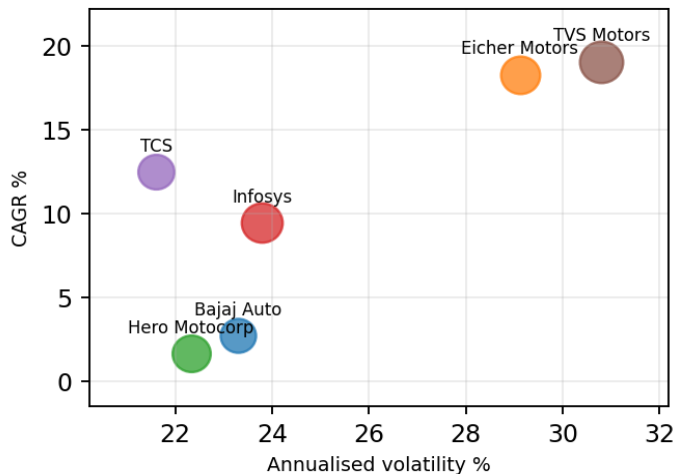
Capital preservation lens: TCS (lowest combined volatility and drawdown rank). Growth lens: TVS Motors. Risk-adjusted lens: TVS Motors. Historical sample only, not a forecast.

# Performance and risk

Growth of 100 invested at the start (bonus-adjusted)



Risk vs return (bubble = max drawdown)



Daily-return correlation

|               | Bajaj Auto | Eicher Motors | Hero Motocorp | Infosys | TCS  | TVS Motors |
|---------------|------------|---------------|---------------|---------|------|------------|
| Bajaj Auto    | 1.00       | 0.33          | 0.52          | 0.16    | 0.16 | 0.36       |
| Eicher Motors | 0.33       | 1.00          | 0.28          | 0.11    | 0.10 | 0.28       |
| Hero Motocorp | 0.52       | 0.28          | 1.00          | 0.18    | 0.18 | 0.33       |
| Infosys       | 0.16       | 0.11          | 0.18          | 1.00    | 0.41 | 0.13       |
| TCS           | 0.16       | 0.10          | 0.18          | 0.41    | 1.00 | 0.11       |
| TVS Motors    | 0.36       | 0.28          | 0.33          | 0.13    | 0.11 | 1.00       |

## Risk and return metrics

|                      | Total return | CAGR   | Volatility | Max drawdown | Sharpe | Sortino | Calmar | VaR 95% (d) | CVaR 95% (d) |
|----------------------|--------------|--------|------------|--------------|--------|---------|--------|-------------|--------------|
| <b>Bajaj Auto</b>    | +10.1%       | +2.7%  | +23.3%     | -23.2%       | -0.14  | -0.21   | 0.12   | -2.0%       | -3.2%        |
| <b>Eicher Motors</b> | +82.6%       | +18.3% | +29.1%     | -28.3%       | 0.42   | 0.67    | 0.65   | -2.7%       | -3.9%        |
| <b>Hero Motocorp</b> | +6.0%        | +1.6%  | +22.3%     | -26.9%       | -0.20  | -0.31   | 0.06   | -2.2%       | -3.0%        |
| <b>Infosys</b>       | +38.2%       | +9.5%  | +23.8%     | -31.1%       | 0.15   | 0.21    | 0.30   | -2.1%       | -3.2%        |
| <b>TCS</b>           | +52.4%       | +12.5% | +21.6%     | -24.1%       | 0.30   | 0.45    | 0.52   | -2.1%       | -3.0%        |
| <b>TVS Motors</b>    | +86.9%       | +19.1% | +30.8%     | -34.6%       | 0.43   | 0.65    | 0.55   | -2.9%       | -4.2%        |

Percentages are period or annualised figures on bonus-adjusted prices; VaR and CVaR are historical daily values. Sharpe and Sortino use the risk-free rate shown on page 1.

# Stock scorecard

0-100 analytical indicators, relative across these stocks. For Volatility, Risk and Drawdown a higher score means more of it. Indicators only, not recommendations.

|               | Momentum | Trend strength | Volatility | Risk (tail loss) | Drawdown | Signal strength | Data quality |
|---------------|----------|----------------|------------|------------------|----------|-----------------|--------------|
| Bajaj Auto    | 17       | 28             | 18         | 18               | 0        | 62              | 100          |
| Eicher Motors | 59       | 44             | 82         | 77               | 45       | 0               | 100          |
| Hero Motocorp | 37       | 36             | 8          | 0                | 33       | 1               | 100          |
| Infosys       | 88       | 89             | 24         | 22               | 69       | 84              | 60           |
| TCS           | 100      | 84             | 0          | 0                | 8        | 100             | 60           |
| TVS Motors    | 0        | 12             | 100        | 100              | 100      | 21              | 100          |

## Data quality

Raw exchange prices contain one-day gaps of around 50% for TCS (31 May 2018) and Infosys (15 Jun 2015). These are 1:1 bonus issues, not crashes. All figures in this report use bonus-adjusted prices so returns, risk and signals are not distorted.

## Method notes

Volatility, CAGR and ratios are annualised on 252 trading days. Signals use the 20/50-day golden-cross rule. The dataset is close-only, so ATR is a proxy and intraday behaviour is not captured.

## Limitations and disclaimer

The sample is small (a handful of stocks over about 3.5 years) and entirely historical. Backtests ignore transaction costs, taxes and slippage. This report is educational analysis and not investment advice.